

- Lenzovo pravilo: sprememba magnetnega pretoka skozi tokokrog požiene tok, ki nasprotuje varovanu nastanka.

- Maxwellova formulacija indukcije

→ Faradayev zakon indukcije: $\Gamma_c = -\frac{d}{dt} \Phi_m$

$$\left. \begin{aligned} \oint_{\partial S} \vec{E} \cdot d\vec{r} &= -\frac{d}{dt} \int_S \vec{B} \cdot d\vec{S} = -\int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S} \\ \int_S \nabla \times \vec{E} \cdot d\vec{S} \end{aligned} \right\} \begin{aligned} \nabla \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t} \\ \text{KINEMATIČNA} \\ \text{MAXWELLOVA} \\ \text{ENAČBA} \end{aligned}$$

- Maxwellov impulz mag. polja

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = -\frac{d}{dt} \nabla \times \vec{A} = -\nabla \times \frac{\partial \vec{A}}{\partial t} \Rightarrow \vec{E} = -\frac{\partial \vec{A}}{\partial t}$$

→ 2. NZ $\vec{E} = -\frac{d}{dt} (e\vec{A}) = \frac{d\vec{p}}{dt}$ ($e\vec{A}$ predstavlja GK, ki jo ob indukciji določamo v sistemu)

- Kvazistatičen sistem Maxwellovih enačb

$$\begin{cases} \nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} & \nabla \cdot \vec{B} = 0 \\ \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} & \nabla \times \vec{B} = \mu_0 \vec{j} \end{cases} \quad \text{Ta sistem (zaradi 4.M.en.) določa } \nabla \cdot \vec{j} = 0, \text{ kar v splošnem ni res}$$

↳ tokarne zanke so zaključene

- Elektromagnetna potenciala za kvazistatična polja

$$\begin{aligned} \nabla \cdot \vec{B} = 0 &\Rightarrow \vec{B} = \nabla \times \vec{A} \\ \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} &= -\nabla \times \frac{\partial \vec{A}}{\partial t} \Rightarrow \nabla \times \left(\vec{E} + \frac{\partial \vec{A}}{\partial t} \right) = 0 \Rightarrow \vec{E} = -\nabla \varphi - \frac{\partial \vec{A}}{\partial t} \end{aligned}$$

→ novi formulaciji veljata v splošnem

- Prevodniki in Ohm zakon

↳ snovi, v katerih 3 prosti nosilci naboja (e^- , vrzeli, ioni, ...)

$$\vec{j} = \sigma \vec{E} \quad \text{Ohm zakon}$$

↳ el. prevodnost

- Časovna konstanta prevodnika (kako hitro se vzpostavi ravnovesje)

$$\nabla \cdot \vec{j} = -\frac{\partial \rho}{\partial t} \rightarrow \nabla \cdot \sigma \vec{E} + \frac{\partial \rho}{\partial t} = 0 \rightarrow \frac{\partial \rho}{\partial t} + \frac{\sigma}{\epsilon_0} \rho = 0 \rightarrow \rho = \rho_0 e^{-\frac{t}{\tau}} \rightarrow \tau = \frac{\epsilon_0}{\sigma} \sim 10^{-18} \text{ s}$$

(kontinuitetna en.)

- Mikroskopski model prevodnosti

→ Drudejev model: $m \frac{d\vec{v}}{dt} = -m \gamma \vec{v} + e\vec{E}$

disipacija (dušenje) sila na nabiti delec

→ ko ni el. polja: $\vec{v} = \vec{v}_0 e^{-\gamma t}$, $W_k = \frac{m v_0^2}{2} e^{-2\gamma t}$ Joulova toplota

→ s poljem ($\vec{E} \neq 0$): $\vec{v}(t) = \frac{e}{m} \int_0^t e^{-\gamma(t-t')} \vec{E}(t') dt'$ $\vec{E} = \text{konst.}$

$$\vec{j} = q n \vec{v} = n e \vec{v} = n e \frac{e}{m} \int_0^t e^{-\gamma(t-t')} \vec{E}(t') dt' = \frac{n e^2}{m \gamma} \vec{E}$$

$$\sigma = \frac{n e^2}{m \gamma}$$

- Električna prevodnost

$$\sigma_0 \left[\frac{S}{m} = \frac{1}{\Omega m} \right] \quad \text{tipično močno odv. od temperature}$$

↑
Siemens

- Upornost

→ el. tok omejimo na vodnik: $\vec{j} = \sigma \vec{E} \rightarrow \int_V \vec{j} \cdot d\vec{r} = \int_V \sigma \vec{E} \cdot d\vec{r} = -\sigma (\varphi(2) - \varphi(1))$

||
 $\int_V \vec{j} \cdot \vec{E} d^3r = I \int_S \frac{dL}{S}$

$$\left. \begin{aligned} \Delta \varphi &= R I, \quad R = \int_S \frac{dL}{\sigma S} \end{aligned} \right\}$$

- Disipacija energije

→ maj. sila deluje ⊥ na smer gibanja, zato ne troši energije

$$\vec{F} = \int q \vec{E} d^3r$$

→ Joulova moč: $P = \int_V \vec{j} \cdot \vec{E} d^3r = \int_V \underbrace{\vec{j} \cdot \vec{E}}_{\text{gostota EM sile}} d^3r = \int_V \underbrace{\vec{j} \cdot \vec{E}}_{\text{gostota sile}} d^3r$ izgube ob gibanju po EMP.

- Kapacitivnost

→ N prevodnikov, $\varphi(\partial V_i) = \text{konst.}$; želimo $C = C U$, $W = \frac{C U^2}{2}$

↑
rob i-tega prevodnika

a) gledamo E celotnega polja: $W_C = \frac{1}{2} \int_V \rho(\vec{r}) \varphi(\vec{r}) d^3r = \frac{1}{2} \sum_i \varphi_i \int_V \rho_i d^3r = \frac{1}{2} \sum_i \varphi_i Q_i$

b) $W_C = \frac{1}{2} \int_V \rho(\vec{r}) \varphi(\vec{r}) d^3r = \frac{1}{2} \int_V \left(\sum_k \frac{Q_k}{4\pi\epsilon_0 |\vec{r} - \vec{r}_k|} \right) \left(\sum_i \frac{Q_i}{4\pi\epsilon_0 |\vec{r} - \vec{r}_i|} \right) d^3r = \frac{1}{2} \sum_{i,k} \frac{Q_i Q_k}{4\pi\epsilon_0 |\vec{r}_i - \vec{r}_k|} = \frac{1}{2} \sum_{i,k} e_i e_k \int_V \frac{1}{|\vec{r}_i - \vec{r}_k|} d^3r = \frac{1}{2} \sum_{i,k} e_i e_k C_{ik}^{-1}$

↑
kapacitivni
samo na
parovih

→ $W_C = \frac{1}{2} \sum_i \varphi_i Q_i = \frac{1}{2} \sum_{i,k} C_{ik}^{-1} e_i e_k$, $C_{ik}^{-1} = \frac{1}{4\pi\epsilon_0 e_i e_k} \int_V \frac{1}{|\vec{r}_i - \vec{r}_k|} d^3r$

- Induktivnost

→ imamo N tokovnih zank, I_i

a) $W_m = \frac{1}{2} \int_V \vec{j} \cdot \vec{A} d^3r = \frac{1}{2} \int_V \vec{I} \cdot \vec{A} d^3r = \frac{1}{2} \sum_i I_i \int_V \vec{A} \cdot d\vec{r}_i = \frac{1}{2} \sum_i I_i \int_V \vec{B} \cdot d\vec{r}_i = \frac{1}{2} \sum_i I_i \Phi_i$

b) $W_m = \frac{1}{2} \int_V \vec{j} \cdot \vec{A} d^3r = \frac{1}{2} \int_V \left(\sum_k \frac{I_k}{4\pi} \frac{\vec{r} - \vec{r}_k}{|\vec{r} - \vec{r}_k|^3} \right) \left(\sum_i \frac{I_i}{4\pi} \frac{\vec{r} - \vec{r}_i}{|\vec{r} - \vec{r}_i|^3} \right) d^3r = \frac{\mu_0}{8\pi} \sum_{i,k} I_i I_k \int_V \frac{1}{|\vec{r}_i - \vec{r}_k|} d^3r$

↑
induk. fr.

→ $W_m = \frac{1}{2} \sum_i I_i \Phi_i = \frac{1}{2} \sum_{i,k} L_{ik} I_i I_k$, $L_{ik} = \frac{\mu_0}{4\pi} \int_V \frac{1}{|\vec{r}_i - \vec{r}_k|} d^3r$

- Kožni pojav: ko izmenični tok teče skozi žico, se razporedi tako, da je gostota toka največja blizu sten prevodnika.

Maxwell + Ohm za prevodnik: $\nabla \cdot \vec{E} = 0$ ($\rho = 0$) $\nabla \cdot \vec{B} = 0$
 $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$ $\nabla \times \vec{B} = \mu_0 \vec{j} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$ (zanemarimo $\epsilon_0 \omega < \sigma$)

$\rightarrow \nabla \times \vec{B} = \mu_0 \vec{j} = \mu_0 \sigma \vec{E}$
 $\left. \begin{aligned} \nabla \times (\nabla \times \vec{E}) &= -\frac{\partial}{\partial t} \nabla \times \vec{B} = -\frac{\partial}{\partial t} \mu_0 \sigma \vec{E} = \nabla (\nabla \cdot \vec{E}) - \nabla^2 \vec{E} \\ \nabla (\nabla \times \vec{B}) &= \mu_0 \sigma \nabla \times \vec{E} = -\mu_0 \sigma \frac{\partial \vec{B}}{\partial t} = \nabla (\nabla \cdot \vec{B}) - \nabla^2 \vec{B} \end{aligned} \right\} \begin{aligned} \nabla^2 \vec{E} &= \mu_0 \sigma \frac{\partial \vec{E}}{\partial t} \rightarrow \text{Nastanek: } \vec{E}(\vec{r}, t) = \vec{E}(\vec{r}) e^{i\omega t} \rightarrow \nabla^2 \vec{E} = k^2 \vec{E} \\ \nabla^2 \vec{B} &= \mu_0 \sigma \frac{\partial \vec{B}}{\partial t} \rightarrow \vec{B}(\vec{r}, t) = \vec{B}(\vec{r}) e^{i\omega t} \rightarrow \nabla^2 \vec{B} = k^2 \vec{B} \end{aligned}$
 $k^2 = -i\omega\mu_0\sigma \rightarrow k = \frac{(1-i)}{\sqrt{2}} \sqrt{\omega\mu_0\sigma} \rightarrow E(\vec{r}, t) \sim e^{-kz} \rightarrow \text{vzorčna globina}$
 $\frac{A}{k} = \frac{\sqrt{2}}{\sqrt{\omega\mu_0\sigma}}$

geometrija polj $\vec{r} \uparrow \int (r, \varphi, z)$
 $\left. \begin{aligned} E_z(r, t) &= E_z(r) e^{i\omega t} \\ B_\varphi(r, t) &= B_\varphi(r) e^{i\omega t} \end{aligned} \right\} \begin{aligned} \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial E_z}{\partial r} \right) &= -i\omega\mu_0\sigma E_z \\ \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial B_\varphi}{\partial r} \right) - \frac{B_\varphi}{r^2} &= -i\omega\mu_0\sigma B_\varphi \end{aligned}$
 $+ \text{vez: } \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \rightarrow -i\omega B_\varphi = -\frac{\partial E_z}{\partial r}$
 $\left. \begin{aligned} E_z &= A J_0(kr) \\ B_\varphi &= -iA \frac{k}{\omega} J_1(kr) \end{aligned} \right\} \rightarrow \vec{j} = \sigma \vec{E} = \hat{e}_\varphi \sigma A J_1(kr)$
 $\rightarrow$ modifinirani Besseli, ker k kompleksen

- glede na velikost ω se loči šibek in močan kožni pojav

